



The Cost of Price: Why and How to Get Beyond Intellectual Property Internalism

Amy Kapczynski

ABSTRACT

The field of intellectual property (IP) law today is focused, as the name itself advertises, on one particular institutional approach to scientific and cultural production: IP. When legal scholars explain this focus, they typically do so with reference to the virtues of price. Because price gives us a decentralized way to link social welfare to the production of information, IP is alleged to be more efficient than other approaches. The dominant mode of IP scholarship begins here and then addresses questions internal to IP law—for example, how broad or narrow should exceptions to IP rights be? But the internalism that characterizes the field of IP cannot, as I show, be justified by the value of efficiency. Economics offers us no a priori reason to assume that IP is more efficient than other possible approaches, most prominently government procurement and commons-based production. If we take the invitation that economists offer us to think external to IP, we also gain new insights about the implications of values other than efficiency for the choice between different institutional approaches to scientific and cultural production. We see, as I argue, that using price to guide scientific and cultural production—which is to say, using IP—may have costs not only for efficiency, but also for distributive justice and informational privacy. The IP approach is in tension with the value of distributive justice because reliance upon price may yield not only unjust distribution of existing information resources but also unjust production of future information resources. The IP approach is in tension with the value of information privacy because relying on price to generate information facilitates the desire, the demand, and perhaps the capacity for price discrimination. That, in turn, generates an impulse for the extensive collection of personal information. Both government procurement and commons-based production plausibly offer more promise than does IP in both distributive justice and privacy terms, and they may be no less efficient than IP. Giving full scope to all three of these values thus requires us to telescope out from the internalism that characterizes the field, and to countenance a broader role for commons-based production and government procurement. In the field of IP, I conclude, we should pay less attention to IP and more to the alternatives.

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TABLE OF CONTENTS

INTRODUCTION	972
I. AN EXTERNALIST VIEW OF EFFICIENCY AND THE AMBIVALENT IMPLICATIONS OF PRICE	981
II. EXTERNALISM, PRICE, AND DISTRIBUTIVE JUSTICE	993
III. INTELLECTUAL PROPERTY EXTERNALISM AND THE COST OF PRICE TO INFORMATIONAL PRIVACY	1006
CONCLUSION	1021

INTRODUCTION

Imagine that instead of a field called “environmental law” we had a field called “cap-and-trade law.” We would, I think, find that odd. Yet this is a fair approximation of where we find ourselves in the field of intellectual property (IP) law today. The field is constructed around one particular institutional approach to sustaining the production of scientific and cultural goods: exclusive rights. Yet it is not at all obvious that IP is categorically superior to other institutional approaches.

Other approaches not only are possible, but also, in many cases, already play an important role in our creative ecosystem. Consider a brief example from the scientific arena. In the United States, about one-third of all research and development (R&D), including more than 80 percent of basic R&D, is funded by government and nonprofit sectors.¹ The results are often disseminated not under property rules, but under norms of open scientific exchange.² This institutional approach,³ which we can call government contracting, is both ubiquitous and familiar. It is employed when government agencies make grants or contract for research, in the manner commonly done by the U.S. National Institutes of Health (NIH) or by the U.S. Department of Defense.

In the cultural arena, the explosion of user-generated content on the internet has fundamentally changed our leisure and entertainment culture. One way to mark this phenomenon is to track the exponential growth of user-generated content sites. Today, more than three billion videos are viewed each day on YouTube, and more video is uploaded to the site in thirty days than was created by the top three U.S. television networks in sixty years.⁴ While some YouTube content is generated by the old-fashioned content industry and is monetized, a substantial portion of it—from parkour performances to mashups to the ubiquitous toddler videos—is generated and shared without any expectation of direct

1. MARK BOROUSH, NAT'L SCI. FOUND., NEW NSF ESTIMATES INDICATE THAT U.S. R&D SPENDING CONTINUED TO GROW IN 2008, at 6 tbl.3 (2010) (2008 data), available at <http://www.nsf.gov/statistics/infbrief/nsf10312/nsf10312.pdf>.
2. See, e.g., J.H. Reichman & Paul F. Uhler, *A Contractually Reconstructed Research Commons for Scientific Data in a Highly Protectionist Intellectual Property Environment*, 66 LAW & CONTEMP. PROBS. 315, 331–32 (2003) (discussing federal policy regarding the open sharing of data stemming from publicly funded research).
3. I call these different “institutional approaches” rather than, say, “alternative mechanisms” because they are not always alternatives but rather are at times complementary and because the word “institution” invokes a thicker and richer field than does the concept of a mechanism.
4. See *Statistics*, YOUTUBE, http://www.youtube.com/t/press_statistics (last visited Feb. 27, 2012).

profit.⁵ We can call this the “commons-based approach” to cultural production, one exemplified by phenomena such as Wikipedia and free software, and one that we can loosely think of as being characterized by nonproprietary norms and by a collaborative, self-organized production process.⁶

A third approach involves financial inducement prizes. Here, a government offers a financial reward to anyone who creates a desired invention—say, a vaccine. The inventor enjoys the benefit of the reward, and the government puts the information it has purchased in the public domain.⁷ This approach is relatively uncommon today, but it has successful historical antecedents, the most famous perhaps being the eighteenth-century British longitude prize.⁸ The prize approach has also generated substantial—and growing—interest in the field of information economics today.⁹

5. YouTube does not offer precise breakdowns of the makeup of its content, but a simple perusal of the site shows that much of the content on YouTube is homegrown. A rough sense of the importance of content on YouTube that is produced without commercial motivation can perhaps be discerned from YouTube’s publicly available figures on monetization. Through its ContentID system, YouTube allows copyright owners to monetize works posted on YouTube, and the site reports monetizing over three billion video views weekly. *Id.* That is just one-seventh of the overall video views, which amount to three billion daily. *Id.* Even assuming that much content-industry material is not monetized, this suggests that a substantial portion—perhaps the majority—of the content viewed on YouTube is not produced through conventional copyright incentives.
6. In the IP literature, the term “commons” is used to refer to a family of approaches that do not always align neatly under a single definition. I discuss this, and offer a more elaborate characterization of the approach, in Part I.
7. Many prize proposals today are structured as supplements to IP incentives and so do not require that the resulting information be put immediately into the public domain. They may, however, impose constraints on pricing or require open licensing under certain conditions. *See, e.g.,* Owen Barder, Michael Kremer & Heidi Williams, *Advance Market Commitments: A Policy to Stimulate Investment in Vaccines for Neglected Diseases*, *ECONOMISTS’ VOICE*, Feb. 2006, at 3, 4 (describing an Advanced Market Commitment (AMC) prize proposal, which obligates medicine manufacturers to sell their products at “low, affordable price[s] in the long-term, or to license their technology to other manufacturers” after sponsors of the AMC have purchased an initial, predetermined number of treatments at higher prices). The classic form of a prize, however, as discussed in Brian Wright’s foundational work, conceives of the prize as a reward in lieu of exclusive rights, thus avoiding the deadweight loss associated with pricing information. Brian D. Wright, *The Economics of Invention Incentives: Patents, Prizes, and Research Contracts*, 73 *AM. ECON. REV.* 691, 691, 697–98, 701 (1983). Prizes of this sort can be structured as IP “buyouts” in particular cases and thus may not disturb the IP system as a whole.
8. *See, e.g.,* DAVA SOBEL, *LONGITUDE* 51–60 (1995).
9. For evidence of the renewed interest in prizes, see Michael Kremer & Heidi Williams, *Incentivizing Innovation: Adding to the Tool Kit*, 10 *INNOVATION POLY & ECON.* 1 (2010); *And the Winner Is . . . Offering a Cash Prize to Encourage Innovation Is All the Rage*, *ECONOMIST*, Aug. 5, 2010, <http://www.economist.com/node/16740639>; Joseph E. Stiglitz, *Prizes, Not Patents*, *PROJECT SYNDICATE* (Mar. 6, 2007), <http://www.project-syndicate.org/commentary/stiglitz/81/English>.

In fact, since the 1960s, economists have debated the relative costs and benefits of a variety of approaches other than IP, most commonly government contracts and financial prizes (which I refer to together as modes of government procurement). In a foundational piece, Kenneth Arrow argued that the most efficient means of producing information likely relied not on property but rather on government.¹⁰ His argument was driven by his analysis—now widely accepted—of the necessary inefficiencies that accompany exclusive rights in information.

Because information is nonrival, as Arrow explained, it has a marginal cost of zero. IP as a strategy works by putting a positive price on information: Creators create, then they charge others to use the information they have produced, thus pricing it above marginal cost. The result is deadweight loss.¹¹ (It is important to note here that economists use the term “information” to refer to an extremely broad range of immaterial resources, from scientific formulas to improvements in production processes to musical compositions to the sequence of words in a novel. These are all products of human minds, and insofar as they are immaterial, they can be shared infinitely without reducing the amount available to others.¹² I follow this usage here since it is common and corresponds to the economic analysis that is central to the field. But I also use the concept of “scientific and cultural production,” in part to invoke a focus on the processes of production rather than their reified result, and in part because the phrase calls forth a wider range of institutional settings than the concept of information production may do.)

A few years later, Harold Demsetz offered an influential response to Arrow. He suggested that despite the deadweight loss that accompanies pricing, property rights in information may be more efficient than government production, largely because of the way that exclusive rights in information guide decisions about the allocation of inventive resources.¹³ Instead of relying on government officials to guide inventive effort, property rights harness the power

10. Kenneth J. Arrow, *Economic Welfare and the Allocation of Resources for Invention*, in THE RATE AND DIRECTION OF INVENTIVE ACTIVITY: ECONOMIC AND SOCIAL FACTORS 609, 623 (Richard Nelson ed., 1962). I discuss the argument in more detail in Part I.

11. One can think of it colloquially in this way: Once a song is produced, you and I and everyone we know can listen to it as many times as we like, with no depletion of the resource. So, ideally, the song would be costless, because any positive price will reduce the access we have to the song and thus reduce welfare. See also *infra* Part I.

12. Transmission could have a positive marginal cost, as will the material goods in which these intangible objects are embedded (such as machines, CDs, books, and the like). But the informational component is nonrival, and economists analyze this informational component separately.

13. Harold Demsetz, *Information and Efficiency: Another Viewpoint*, 12 J.L. & ECON. 1, 14 (1969).

of price to transmit information between consumers and decentralized creators.¹⁴ Price links the production of information to consumer demand, and, by extension, to social welfare.

This argument has been so deeply internalized in the field of IP law that it is typically taken for granted.¹⁵ For example, the introduction of the leading IP casebook, coauthored by Robert Merges, Peter Menell, and Mark Lemley, states decisively that “[i]ntellectual property protection is necessary to encourage inventors, authors, and artists to invest in the process of creation.”¹⁶ It justifies the institutional preference for IP only briefly, with a reference to the virtues of decentralized market-based systems over “central planning on a mass scale.”¹⁷ This reflects the dominant view in the field of IP law today, which—as is advertised in the name—focuses on intellectual property rights as the privileged means of promoting scientific and cultural production. The choice is justified, if at all, with reference to Demsetzian arguments that are said to establish IP as more efficient than the alternatives. The core of the field of IP law today thus focuses on questions “internal” to IP—for example, on questions about the effects or interpretation of a particular doctrine of copyright or patent law, about how broad or narrow the scope of exclusive rights or their exceptions should be, or about the administration of the IP system and how it can be improved. The effect is to focus our attention on questions about tradeoffs within IP law, and to suggest that the stakes of these questions are more important than the stakes of questions about the trade-offs between IP and other institutional approaches to cultural and scientific production.

There is also a long and important tradition of dissent from the dominant internalism of the legal field.¹⁸ Over the past few years, that dissent has gained

14. *Id.* at 12–13. By “price” here, I simply mean payment voluntarily exchanged in private markets. Information is priced, in the sense I mean here, when it is treated as a commodity using a property rule.

15. For examples, see *infra* note 40.

16. ROBERT P. MERGES, PETER S. MENELL & MARK A. LEMLEY, *INTELLECTUAL PROPERTY IN THE NEW TECHNOLOGICAL AGE* 14 (5th ed. 2010).

17. *Id.* at 18. The authors go on to note that the case for IP is based “more on a generalized perception of institutional choice than on strong direct evidence of the superiority of intellectual property rights relative to the alternatives.” *Id.* But they offer no further discussion of the potential tradeoffs between different institutional choices, or of the robust economics literature that calls into question a generalized preference for IP.

18. For an early and prominent dissent to the assumption that IP is necessary for economic efficiency, see Stephen Breyer, *The Uneasy Case for Copyright: The Study of Copyright in Books, Photocopies, and Computer Programs*, 84 HARV. L. REV. 281, 283–84, 293–308 (1970) (concluding that the economic case for copyright in books “is not a strong one,” arguing that market forces such as lead time may be sufficient). For a later work that considers a broader range of alternatives to IP,

in prominence. A small but growing number of legal scholars have begun to argue for the potential of prizes and other modes of government procurement as alternatives or complements to an IP-based approach.¹⁹ There has also been a surge of interest among legal scholars in commons-based production of scientific and cultural goods.²⁰ There is also a notable policy

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- see Steve P. Calandrillo, *Economic Analysis of Property Rights in Information: Justifications and Problems of Exclusive Rights, Incentives to Generate Information, and the Alternative of a Government-Run Reward System*, 9 FORDHAM INTELL. PROP. MEDIA & ENT. L.J. 301 (1998). Early work by Rebecca Eisenberg and Arti Rai focused attention on academic and government-supported scientific research, for example, by analyzing the relationship between the norms that govern that realm and IP norms. See, e.g., Rebecca S. Eisenberg, *Proprietary Rights and the Norms of Science in Biotechnology Research*, 97 YALE L.J. 177 (1987); Arti Kaur Rai, *Regulating Scientific Research: Intellectual Property Rights and the Norms of Science*, 94 NW. U. L. REV. 77 (1999). William Fisher has sought to bring alternatives to IP more firmly to the center of IP conversations, both through theoretical arguments, see William W. Fisher, *Intellectual Property and Innovation: Theoretical, Empirical and Historical Perspectives*, in INDUSTRIAL PROPERTY, INNOVATION, AND THE KNOWLEDGE-BASED ECONOMY *2-4 (A. Arundel ed., 2002), available at <http://cyber.law.harvard.edu/people/ffisher/Innovation.pdf>, and through detailed policy proposals, see WILLIAM W. FISHER III, PROMISES TO KEEP: TECHNOLOGY, LAW, AND THE FUTURE OF ENTERTAINMENT (2004). Yochai Benkler has made similarly important contributions, particularly with his book on the phenomenon of commons-based peer production, YOCHAI BENKLER, THE WEALTH OF NETWORKS: HOW SOCIAL PRODUCTION TRANSFORMS MARKETS AND FREEDOM (2006). The rich literature on the public domain also has helped map the "outside" of IP, although not always with a specific focus on the viability of other institutional approaches to scientific and cultural production. For key contributions on the public domain, see JAMES BOYLE, THE PUBLIC DOMAIN: ENCLOSING THE COMMONS OF THE MIND (2008); David Lange, *Recognizing the Public Domain*, 44 LAW & CONTEMP. PROBS. 147 (1981); Jessica Litman, *The Public Domain*, 39 EMORY L.J. 965 (1990); Pamela Samuelson, *Enriching Discourse on Public Domains*, 55 DUKE L.J. 783 (2006).
19. For recent legal scholarship on the potential of prizes, see, for example, FISHER, *supra* note 18; Michael Abramowicz, *Perfecting Patent Prizes*, 56 VAND. L. REV. 115 (2003); Jonathan H. Adler, *Eyes on a Climate Prize: Rewarding Energy Innovation to Achieve Climate Stabilization*, 35 HARV. ENVTL. L. REV. 1 (2011); Steven Shavell & Tanguy van Ypersele, *Rewards Versus Intellectual Property Rights*, 44 J.L. & ECON. 525 (2001); see also Neil Weinstock Netanel, *Impose a Noncommercial Use Levy to Allow Free Peer-to-Peer File Sharing*, 17 HARV. J.L. & TECH. 1 (2003) (promoting a variant of a prize proposal). The potential of government contracting has received less attention, but has been treated as promising by some legal scholars. See Jerome H. Reichman, *Rethinking the Role of Clinical Trial Data in International Intellectual Property Law: The Case for a Public Goods Approach*, 13 MARQ. INTELL. PROP. L. REV. 1 (2009).
 20. See, e.g., BOYLE, *supra* note 18; Yochai Benkler, *Freedom in the Commons: Towards a Political Economy of Information*, 52 DUKE L.J. 1245 (2003); Jorge L. Contreras, *Data Sharing, Latency Variables, and Science Commons*, 25 BERKELEY TECH. L.J. 1601 (2010); Peter Lee, *Toward a Distributed Commons in Patent Law*, 2009 WIS. L. REV. 917; Michael J. Madison, Brett M. Frischmann & Katherine J. Strandburg, *Constructing Commons in the Cultural Environment*, 95 CORNELL L. REV. 657 (2010); Stephen M. Maurer, Arti Rai & Andrej Salj, *Finding Cures for Tropical Diseases: Is Open Source an Answer?*, 6 MINN. J.L. SCI. & TECH. 169 (2004); Yochai Benkler, *The Commons as a Neglected Factor of Information Policy* (Working Draft Presented at Telecomms. Pol'y Research Conference, 1998), available at <http://www.benkler.org/commons.pdf>. See also the many commentaries on the Madison, Frischmann, and Strandburg article in the

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interest in alternatives to IP approaches today, particularly in the potential of prizes.²¹

I seek here to build on this dissent, by showing first that IP internalism cannot, as a general matter, be justified by appeals to efficiency. That is because, as Part I shows, the contemporary field of information economics itself offers no clear endorsement of IP. That literature has moved beyond Demsetz's response to Arrow and has developed a complex conversation about the comparative costs and benefits of prizes, contracts, and IP—a conversation that has produced no definitive resolution in favor of any of these approaches. This is well understood by those in the IP field who reject the internalist impulse.²² Scholars have also called attention to commons-based production as another approach that may be as or more efficient than IP. The existing legal and economic literature teaches us, then, the question of which institutional approach to information production is most efficient must be examined in context, and in any case it may be difficult to determine with any certainty.

If concerns about efficiency do not dictate a preference for IP, might other fundamental values have something to say about the matter? As IP scholars have pointed out, the choices we make about information governance implicate a range of fundamental values, including values of privacy and distributive justice.²³ Discussions about these two values have also tended to follow an internalist frame, asking, for example, whether distributive values generate arguments for

Cornell Law Review, volume 95. This literature is still in its infancy, and the definition of "commons-based production" is contested (and is discussed further in Part I). But one measure of its importance is the fact that a critical counter-literature has begun to emerge. See, e.g., Jonathan M. Barnett, *The Illusion of the Commons*, 25 BERKELEY TECH. L.J. 1751 (2010).

21. See, e.g., COMMITTEE ON THE DESIGN OF AN NSF INNOVATION PRIZE, NAT'L RESEARCH COUNCIL, INNOVATION INDUCEMENT PRIZES AT THE NATIONAL SCIENCE FOUNDATION (2007); *Centennial Challenges*, NASA, http://www.nasa.gov/offices/oct/early_stage_innovation/centennial_challenges/index.html (last visited Feb. 27, 2012).

22. Benkler and Fisher have done particularly important work to elucidate this point. See BENKLER, *supra* note 18; FISHER, *supra* note 18.

23. See, e.g., Margaret Chon, *Intellectual Property and the Development Divide*, 27 CARDOZO L. REV. 2821, 2907 (2006) (describing the potential conflicts between IP law and distributive justice, particularly in relation to development); Julie E. Cohen, *DRM and Privacy*, 18 BERKELEY TECH L.J. 575 (2003) [hereinafter Cohen, *DRM and Privacy*] (discussing the conflicts between digital rights management (DRM) technologies, which are part of the enforcement infrastructure for copyright, and privacy values). Scholars invoke a range of other values as well, of course, such as democracy, authorship, creativity, freedom of thought or cultural engagement, free speech, and the like. No single article could analyze the implications of all such values for the choice of institutional approaches to scientific and cultural production. I have focused on the values of distributive justice and privacy here to illustrate the implications of the externalist approach, and because they offer us an entry point to important new arguments about the tradeoffs between different institutional approaches to innovation.

exceptions to existing IP law. If we adopt an externalist perspective, instead, we bring into view something otherwise obscured. Using price to guide scientific and cultural production—which is to say, using IP—may itself generate costs from the perspective not only of efficiency, but also of distributive justice and privacy.

In Part II, I show that the institutional approach of IP is in significant tension with distributive values. The most obvious reason, which is in some sense well known and visible even from an internalist perspective, is that IP uses price to ration access to information goods. Price can be a problematic way of distributing goods that are important to justice because the existing distribution of resources may be unjust. While this concern also exists outside the domain of information, it is particularly meaningful here, because here we know that price is not necessarily a more efficient means of production than are other institutional approaches. The usual response to distributive concerns about price, which suggests that we should use price to maximize efficiency and redistribute afterwards, therefore has little purchase in the context of information.

The more pertinent and common objection to distributive impulses internal to IP law is that yielding to such impulses will undermine their own aim. In other words, limiting the reach of IP will undermine the production of the very information that must be fairly distributed due to its importance to justice. When scholars debate the implications of distributive justice for international patent law from an internalist perspective, for example, they risk being caught in an eddy: The desire to make sophisticated medicines available to the poor around the world seems to require not only thin patent law (so that affordable generic drugs are available) but also robust patent law (so that new medicines for the poor are created).²⁴ From inside of IP, it is difficult to dissolve this apparent paradox. As I show, it can be dissolved because sometimes we can make modifications to IP law that will have effects that are more beneficial than harmful for the poor. But the results in distributive terms will be modest, leaving a substantial domain of inequality that cannot be addressed from within an internalist paradigm.

If we adopt an externalist view, distributive justice arguments have a much broader scope. As I show, from a distributive perspective, IP is not just a problematic way to distribute cultural and scientific goods that are important to justice, but it is also a problematic way to produce them. In an IP system, price influences not only who has access to such goods, but also which goods are produced in the first place. As long as the rich and the poor sometimes have different needs,

24. For an example, see *infra* text accompanying notes 93–94.

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as they demonstrably do, a system based on price will prioritize the needs of the wealthy. Moreover, some information goods may be of such foundational importance to human freedoms and capabilities that all should have them regardless of their preferences. Because price necessarily tracks preferences, it is a particularly problematic way to develop and distribute these kinds of basic information goods. Commons-based production and particularly government procurement plausibly have advantages over IP in distributive terms because they yield information goods that are made available for free, and are better suited to attempts to direct production in ways that are divorced from preferences and from the existing allocation of resources.

Part III addresses the implications of our choice of institutional approach for informational privacy. As I argue, and as an externalist view allows us to see, choosing IP as the mode for producing and distributing information sets up a constitutive tension with information privacy. When we produce and distribute information through price we have the problem of deadweight loss. The solution proffered is price discrimination, in which a producer charges a user a specifically tailored price, matched to her individual willingness and ability to pay.²⁵ Price discrimination is generally thought to enhance efficiency in the presence of price.²⁶ Moreover, when we rely on price, we generate a decentralized impulse to price discriminate because price discrimination will enhance the profits of producers. But price discrimination and privacy are in tension with one another because personal information is critical to the ability to cheaply and accurately price discriminate. While government procurement and commons-based schemes may also trigger privacy concerns, there are reasons to think that these concerns may be less acute than those that accompany price—and thus IP.

In this Article, I thus offer two primary contributions to the existing literature. First, I identify the tendency toward IP internalism that characterizes the field today and demonstrate that it is not justified by efficiency arguments. Giving full scope to efficiency values would require us instead to adopt an externalist approach—to telescope out from a singular focus on IP as the privileged way to promote scientific and cultural production and to bring other institutional approaches from the periphery to the center of our inquiry. Second, I

25. A more formal definition of price discrimination is the state of affairs “when the prices of similar products sold by the same firm show variation that cannot be attributed to cost variation.” Frank Verboven, *Price Discrimination (Empirical Studies)*, in *THE NEW PALGRAVE DICTIONARY OF ECONOMICS* (Steven N. Durlauf & Lawrence E. Blum eds., 2d ed. 2008).

26. This argument is described in detail below. *See infra* note 39.

argue that we also cannot give full scope to values beyond efficiency without adopting an externalist viewpoint. More specifically, I offer new arguments about the tension between the institutional approach of IP and values of distributive justice and informational privacy, and seek to show that both government procurement and commons-based approaches plausibly may generate less tension with both values.

My aim is not to argue that a particular institutional approach or mix of institutional approaches is always best. Indeed, my sense is that the contexts and resources involved are too diverse to admit to resolution in any such general way. Rather, I contend that we should systematically expand our field of vision so that the tradeoffs and complementarities between IP and other approaches become a central subject of the field, and so that we can more clearly debate the implications of our choice of approach not only for values of efficiency, but also for values such as distributive justice and privacy. Notably, an external approach not only gives more scope to our normative inquiries, but also better corresponds to the fundamentally mixed nature of our existing information ecology.

Renaming the field to encompass something broader than IP as an approach would facilitate the kind of analysis called for in this Article. It is worth noting in this regard that the term "intellectual property" only recently came into prominence.²⁷ But in calling for IP externalism, I mean also to suggest that we do not need to rename the field in order to give it a new geography. Property law has been reinvented many times, all under the same name. We can telescope out and adopt a broader frame of reference for our inquiries, regardless of what we call our efforts. Indeed, this move is already underway. If we embrace it as a priority, a new field of analysis opens up.

27. For a review of the evidence about the emergence of the term, see Amy Kapczynski, *The Access to Knowledge Mobilization and the New Politics of Intellectual Property*, 117 YALE L.J. 804, 843, 846–47 (2008). For an influential account of the rise of the term, see Mark A. Lemley, *Property, Intellectual Property, and Free Riding*, 83 TEX. L. REV. 1031, 1033–34 (2005) (charting an “exponential” rise in the use of the term “intellectual property” in federal court opinions, with the inflection point in the 1980s). What is new is less the proprietization of IP than its intellectualization—that is, the grouping of different doctrinal and statutory regimes under this collective rubric. See, e.g., Peter K. Yu, *Intellectual Property and the Information Ecosystem*, 2005 MICH. ST. L. REV. 1, 4.

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because of the problems of transaction costs, externalities, and racing.⁸⁶ Additionally, the price mechanism can distort allocation away from the optimum if it provides more than competitive returns in an environment in which normal commodities only produce competitive returns.⁸⁷

In sum, there are many modes of producing information. As old debates and recent developments in the field of information economics show, no general case can be made that the price mechanism is the most efficient means of producing and disseminating information. The seriousness of transaction cost problems in the context of information, as well as the existence of commons-based strategies of information production, provide further challenges to any assumption that price is the best way to efficiently produce information goods. Whether price, and thus IP, will serve as a better or worse form of inducement than will the many other possibilities depends on a multitude of factors that, at a minimum, must be assessed in context, and that may be difficult to assess with a high degree of confidence.

Paradoxically, then, although IP law is typically defended with reference to values of efficiency, information economics teaches us that there is no reason to treat IP as clearly more efficient than the alternatives. This is readily apparent if we begin not from an internalist perspective, but rather from an externalist perspective and from the broader frame of reference that it permits.

II. EXTERNALISM, PRICE, AND DISTRIBUTIVE JUSTICE

Although IP scholars typically reason in the idiom of efficiency, a small but growing number of them have begun to suggest that distributive justice values should also influence information policy. But almost invariably, they do so from an internalist perspective. An externalist perspective gives far more scope to distributive concerns. It also allows us to see new dimensions of the

86. These are not the only objections to this argument. Also important are possible dynamics of crowding out of inventive effort when payment is deployed. See *infra* note 198. The distributive problems of markets are also relevant here because they suggest that price may systematically diverge from social welfare when the poor are limited not by their willingness but rather by their ability to pay. See *infra* Part II.

87. See Calandrillo, *supra* note 18, at 330; Lunney, *supra* note 49, at 577 ("If copyright expands its protection too much in an attempt to 'award' an author the 'value' of her work when the rest of our economy operates on a 'cost' basis, it will generate both inefficiency and unfairness."). Moreover, if the IP mechanisms used to harness the price mechanism are also incomplete, in the sense that they do not encompass all forms of valuable information, this will yield another kind of allocative distortion. See Amy Kapczynski & Talha Syed, Nonexcludability and the Limits of Patents (Feb. 10, 2012) (unpublished manuscript) (on file with author).

problem that IP can pose for distributive justice, and the concomitant potential of government procurement and commons-based approaches from a distributive perspective.⁸⁸

Molly Van Houweling's analysis of the implications of copyright law's design for poor creators is one of the best developed distributive justice arguments in IP scholarship, and it offers a good example of the internalist approach.⁸⁹ She is concerned with the distribution of creative opportunity, and she therefore focuses on the burdens that copyright law may impose upon creators who have little money.⁹⁰ She argues that the problems copyright may pose from a distributive perspective have become more acute as new digital technologies have democratized creation and have made amateurs more visible to those who want to extract revenues from the copyrighted works that amateurs may draw upon.⁹¹ She then proposes a shift in copyright law so that those who are not profiting from their creations could qualify more easily for the fair use defense.⁹²

Another substantial attempt to introduce distributive values to IP, this time in the context of international IP law, comes from Margaret Chon. Her analysis also proceeds from an internalist perspective, focusing on the constraints that international treaties impose on the distribution of human capabilities around the world, as well as offering policy solutions internal to that law (for example, the

88. Others have called attention to parts of this proverbial elephant, and I draw on their work below. Benkler has argued for the potential distributive benefits of the commons. See *infra* text accompanying note 114. Fisher's argument that a prize system is structurally better suited than the copyright system to promote semiotic democracy seems to embody a distributive concern, and it depends on insights about the nature of exclusive rights systems that are similar to those discussed below. See FISHER, *supra* note 18, at 28–31. Fisher and Syed have argued that prizes are better suited than patents to address the problem of neglected diseases in developing countries, apparently with distributive concerns in mind. See Fisher & Syed, *supra* note 65, at 9. Although not a legal scholar, James Love's work in the area of neglected diseases has distributive concerns at its core and has been important to the thinking of many legal scholars on distributive questions in IP, including myself. For examples of his work, see James Love, *Prizes to Stimulate Innovation*, KNOWLEDGE ECOLOGY INT'L (Aug. 12, 2009, 10:12 AM), <http://keionline.org/prizes>.

89. See Molly Shaffer Van Houweling, *Distributive Values in Copyright*, 83 TEX. L. REV. 1535 (2005).

90. *Id.* at 1537.

91. *Id.* at 1539.

92. *Id.* at 1567. Fair use doctrine permits unauthorized copying of some or all of a copyrighted work, and depends on considerations such as whether the copying is transformative and the effect that the copying might have on the market for the original work. For insightful analysis of how fair use is applied by courts, see Barton Beebe, *An Empirical Study of U.S. Copyright Fair Use Opinions, 1978–2005*, 156 U. PA. L. REV. 549 (2008); Pamela Samuelson, *Unbundling Fair Uses*, 77 FORDHAM L. REV. 2537 (2009).

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expansion of the exceptions and limitations in IP treaties).⁹³ The common rejoinder to such proposals is that increasing flexibilities in international law will undermine incentives to innovate.⁹⁴ To this, Chon says that “equality tilts the balance towards static efficiency and away from dynamic efficiency arguments, at least for resource-poor areas of the world.”⁹⁵ As this statement exemplifies, from an internalist perspective, those concerned with distributive justice appear to be caught in a paradox that requires them to choose between access to information and the creation of information. This is in fact not strictly so, but it is the case that for debates about information policy, taking an internalist viewpoint truncates the implications of a distributive justice perspective.

The literature on distributive justice in the IP field is meager enough that it is worth beginning by clarifying just what such an approach implies. Theories of distributive justice address the just allocation of resources in a society.⁹⁶ While theorists of distributive justice rarely make reference to information itself as a resource of concern,⁹⁷ it is without doubt that information is of substantial importance to distributive justice today. For example, textbooks, newspapers, and medicines are all informational goods (by which I mean goods whose value is in important part constituted by their information content). Art, literature, and music are also encompassed under the broad rubric of information, and they are of clear importance to community and political life.

93. See, e.g., Chon, *supra* note 23, at 2907. Chon draws upon the capabilities framework of Amartya Sen, which is described later in this Part. Other notable arguments in the distributive justice vein include one offered by Syed, who focuses on the temporal problem that IP systems will generate innovations for which the poor must wait to access. See Talha Syed's Remarks at the A2K4 Conference on Access to Knowledge and Human Rights at Yale Law School (Feb. 12, 2010, 11:06 AM), <http://www.ustream.tv/recorded/4670860>. But this, too, remains within an internalist frame.

94. See, e.g., Richard A. Epstein & F. Scott Kieff, *Questioning the Frequency and Wisdom of Compulsory Licensing for Pharmaceutical Patents*, 78 U. CHI. L. REV. 71, 80 (2011); Alan O. Sykes, *TRIPS, Pharmaceuticals, Developing Countries, and the Doha "Solution,"* 3 CHI. J. INT'L L. 47, 49 (2002).

95. Chon, *supra* note 23, at 2891.

96. See, e.g., G.A. Cohen, *Where the Action Is: On the Site of Distributive Justice*, 26 PHIL. & PUB. AFF. 3, 3 (1997) (stating that distributive justice refers to the “just distribution of benefits and burdens in society”); see also Serge-Christophe Kolm, *Distributive Justice*, in A COMPANION TO CONTEMPORARY POLITICAL PHILOSOPHY 438, 438–39 (Robert E. Goodin & Philip Pettit eds., 1995) (defining distributive justice and noting that theories of distributive justice differ according to the “ethically relevant variables” that they deem to be important to the just allocation of resources).

97. There are exceptions to this statement, although, to my knowledge, references to information itself as a resource of concern tend to be episodic. Sen, for example, specifically cites “the reform of patent laws to make well-established and cheaply producible drugs more easily available to needy but poor patients (for example, those who are suffering from AIDS)” as a question of justice that a capabilities approach can address, but he does not explain exactly how the capabilities approach applies to this context. AMARTYA SEN, *THE IDEA OF JUSTICE* 24–25 (2009).

The intensity of informational goods' importance to well-being is perhaps easiest to illustrate in the domain of health. Since the 1950s, global average life expectancy has increased by almost twenty years.⁹⁸ According to the World Health Organization, these gains are largely attributable to "[t]he application of knowledge from health research"⁹⁹ to improve, for example, sanitation and access to vaccines. These gains are, of course, unevenly distributed, and up to ten million lives per year could be saved simply by providing better access to existing informational goods such as medicines and vaccines.¹⁰⁰ More research aimed at developing new vaccines and medicines for diseases that particularly affect the poor in developing countries could save many more lives still.¹⁰¹

IP is problematic, from a distributive justice perspective, for reasons that in some sense are well known: IP rations access via the price mechanism, and so it distributes resources in a way that is sensitive to the background allocation of resources. Yet the background allocation of resources may be unjust.¹⁰² This is the implicit basis of the internalist arguments made by both Chon and Van Houweling, for example: Copyright and patent law both disadvantage the virtuous poor because they make price a ticket for entry. Conversely, the poor will benefit disproportionately, in terms of access, when information goods are available without price or at a low price.

This basic tension between price and access to goods that are important to justice is familiar, and it is not limited to the domain of information. The usual response from those who favor price is that efficiency should be maximized first, and that distributive issues should be attended to thereafter.¹⁰³ The argument

98. WORLD HEALTH ORG., THE WORLD HEALTH REPORT 2003: SHAPING THE FUTURE 1 (2003).

99. WORLD HEALTH ORG., WORLD REPORT ON KNOWLEDGE FOR BETTER HEALTH: STRENGTHENING HEALTH SYSTEMS 1 (2004).

100. See WORLD HEALTH ORG., EQUITABLE ACCESS TO ESSENTIAL MEDICINES: A FRAMEWORK FOR COLLECTIVE ACTION 1 (2004) (concluding that ten million lives could be saved each year by increasing access to existing medicines).

101. See, e.g., William W. Fisher & Talha Syed, *Global Justice in Healthcare: Developing Drugs for the Developing World*, 40 U.C. DAVIS L. REV. 581, 612 (2007) (arguing that expenditures for research on neglected diseases should be increased from the current 2 to 3 percent of global medical research expenditures to reflect the fact that neglected diseases account for 16.4 percent of the global disease burden).

102. Many theorists of distributive justice make this point, of course. For one argument that an initial just allocation of minimum resources is a condition of distributive justice in the liberal state, see BRUCE ACKERMAN & ANNE ALSTOTT, THE STAKEHOLDER SOCIETY (1999).

103. For the foundational elaboration of this argument, see Louis Kaplow & Steven Shavell, *Why the Legal System Is Less Efficient Than the Income Tax in Redistributing Income*, 23 J. LEGAL STUD. 667, 674-75 (1994).

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has intuitive appeal; if one wants to redistribute resources, one must have resources to redistribute. This argument is not without its detractors,¹⁰⁴ and in particular it applies poorly beyond the nation-state, where no post hoc tax and transfer mechanisms exist. But whatever the general merits of this argument may be, it has little purchase in the context of information goods. As described above, in the context of information, the effects of price on efficiency are uncertain. If we can get the same or similar results in terms of efficiency, but intrinsically better distributive results (avoiding, for example, both the cost and the political uncertainty of post hoc redistribution), then there is no reason to accept the cost (in distributive terms) that price requires.

The problem with distributive justice arguments that favor IP exceptions is less this familiar argument that we should prioritize efficiency than it is the apparently self-defeating nature of these exceptions. Because IP both helps to create and constrains access to the goods that we want, we have the appearance of a paradox: If we make IP more flexible so that the poor have more access, the poor will also have less access—because the information that we want the poor to access is less likely to be created. This dilemma causes Chon, for example, to assert that static concerns (namely, access today) must be prioritized over dynamic ones (namely, future scientific and cultural production).

There is in fact no true paradox here, because sometimes we can give the poor access to information goods without substantially undermining innovation. This is best illustrated by arguments developed in the transnational context showing that developing countries are best off if they make liberal use of their right under international IP law to override patents on medicines.¹⁰⁵

While this move reduces returns to innovators, developing countries make up a very small portion of innovator markets; all of Africa, for example, makes up just

104. See, e.g., Jules L. Coleman, *The Rational Choice Approach to Legal Rules*, 65 CHI.-KENT L. REV. 177, 179 (1989); Christine Jolls, *Behavioral Economics Analysis of Redistributive Legal Rules*, 51 VAND. L. REV. 1653 (1998); Duncan Kennedy, *Law-and-Economics From the Perspective of Critical Legal Studies*, in 2 THE NEW PALGRAVE DICTIONARY OF ECONOMICS AND THE LAW 465, 470–71 (1998); Duncan Kennedy, *Cost-Benefit Analysis of Entitlement Problems: A Critique*, 33 STAN. L. REV. 387, 401 (1981); Richard S. Markovits, *On the Relevance of Economic Efficiency Conclusions*, 29 FLA. ST. U. L. REV. 1, 37–40 (2001).

105. See, e.g., Agreement on Trade-Related Aspects of Intellectual Property Rights art. 31, Apr. 15, 1994, Marrakesh Agreement Establishing the World Trade Organization, Annex 1C, 1869 U.N.T.S. 299, 333; World Trade Organization, Ministerial Declaration of 14 November 2001, WT/MIN(01)/DEC/2, 41 I.L.M. 755 (2002).

1 percent of the global market for innovator pharmaceutical companies.¹⁰⁶ Even when considering all developing countries together and looking at their share of gross domestic product, rather than looking at the existing state of pharmaceutical markets (which are affected by patent regimes), the marginal incentive that could be created by market demand in these jurisdictions is relatively small, while the barrier to access that patents may pose to local consumers is large.¹⁰⁷ It is thus possible to make exceptions to IP law that will disproportionately benefit the poor, despite the potential dynamic effects. The use of so-called compulsory licenses in poor countries to promote access to medicines represents precisely this kind of move.

This point can be generalized: Often the poor (and especially the very poor) are so poor that they make up very little of an expected market for an innovation. Giving them access to goods by exempting them from exclusion rights can thus yield more in access than it would compromise in terms of innovation. Even from an efficiency perspective, and certainly from a distributive perspective, such users should ride for free if the scheme can be administered with reasonable accuracy and without excessive expense.¹⁰⁸

But the free-riding strategy has clear limitations. It may be difficult or costly to identify the poor. And, at some point, access for the poor will undercut incentives substantially enough to undermine incentives to generate information. Again, we can illustrate with an example in the medicines context. It is no accident that the global access to medicines campaign and its focus on addressing patent barriers to medicines arose out of the HIV/AIDS movement. A large enough community of people living with HIV in wealthy countries existed to attract investment into new medicines to treat HIV. No similar interest exists

106. WORLD HEALTH ORG., COMM'N ON INTELLECTUAL PROP. RIGHTS, INNOVATION AND PUB. HEALTH, PUBLIC HEALTH, INNOVATION AND INTELLECTUAL PROPERTY RIGHTS 15 tbl. 1.4 (2006) (2005 data provided by IMS health).

107. Frederick Scherer shows that if extending patents to all low-income nations increased rents by these nations' share of global gross domestic product (20 percent), the innovative effect would be far less than what would be required to offset the projected deadweight loss. F.M. Scherer, *A Note on Global Welfare in Pharmaceutical Patenting*, 27 WORLD ECON. 1127, 1128-29 (2004).

108. The main costs to such a system would be the cost of accurately identifying the poor and the risk of arbitrage. But implementing a norm in favor of free riding by the poor might not be as expensive or difficult as it might seem to be. Reasonably effective and moderately leaky ways to segment rich and poor markets may exist, including targeting institutions that disproportionately serve the poor (such as public schools and public libraries in low-income areas), linking the benefit of free riding to existing means-testing schemes (such as federal student assistance or Medicaid), or using national borders as a rough measure of poverty. We could also imagine less centralized schemes that give information industries tax incentives or that impose regulatory requirements to encourage industry to identify and serve low-income users themselves.

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in developing treatments for conditions such as extensively drug-resistant tubercu-
losis (TB)—the largest impact of which is felt in South Africa.¹⁰⁹ No patent
exception can give patients with this form of TB better access to simple and
fast-acting medicines, because such medicines do not exist.

Fundamentally, the point is this: From an internalist perspective, it is poss-
ible to locate areas in which exceptions to IP law will yield distributive advantages
because they will give the poor much more access and will do little to undermine
innovation. But whether this is so will be a function of the state of the relevant
markets, and it will be orthogonal to the importance of the issue from a distri-
butive justice perspective. From an external perspective, it becomes clear that
while such exceptions may not undermine information production, they can also
do little to support it.

In other words, using IP to generate innovation will undersupply not just
access for the poor but also production for the poor. Yet distributive justice
plausibly demands not only fair distribution of information goods but also
equitable production of information goods.¹¹⁰ Suspending price alone can help
to achieve fairer distribution but not equitable production. The same pattern will
be reproduced by all distributive exemptions in IP law whether they come under
the guise of fair use in the copyright context, the research exemption in patent
law, or exceptions that deliberately target the poor. For example, if justice de-
mands that we allocate equal or similar resources to the development of medical
treatments for both rich and poor people, then using price to drive production
will misfire: Under conditions of limited ability to pay, price will signal that the
lives of the wealthy are more important than are the lives of the poor.

The problem will appear to be the most acute when the rich and the poor
have different information needs, because here the information goods that the

109. See Sheela Shenoi et al., *Multidrug-Resistant and Extensively Drug-Resistant Tuberculosis: Consequences for the Global HIV Community*, 22 CURRENT OPINIONS INFECTIOUS DISEASES 11, 16 n.7 (2009); *Why New TB Drugs? MDR-TB, XDR-TB*, TB ALLIANCE, <http://www.tballiance.org/why/mdr-xdr.php> (last visited Feb. 27, 2012). Because of the very limited ability of the poor in developing countries to pay, very few medicines are developed for diseases that particularly affect them. For example, a recent study revealed that only 1 percent of medications introduced between 1975 and 1999 targeted tuberculosis and tropical diseases. Patrice Trouiller et al., *Drug Development for Neglected Diseases: A Deficient Market and a Public-Health Policy Failure*, 359 LANCET 2188, 2189-90 (2002).

110. This is not to discount possible arguments that we should give some kind of priority to access to existing information, even at the expense of future production. Such arguments invoke difficult questions of interpersonal and intergenerational justice, which are beyond the scope of this project. But it is certainly plausible that the production of information is as much an issue of distributive justice as is access to existing information, which is all that I mean to suggest here.

poor need will not be developed at all. This is an evident problem in health, in which the global poor disproportionately suffer from communicable diseases that have little or no impact in the global North.¹¹¹ But it could also be an issue in the cultural domain if, for example, the poor have needs for distinct educational goods or political literatures.¹¹² There is also a less visible cost even in those areas where the rich and the poor share a need, such as in the area of medicines for noncommunicable diseases. There, relevant information may be produced (say, medicines or techniques to reduce blood pressure), but produced more slowly or in a smaller supply than it would be if the poor had the same ability as do the rich to make their demands felt in the marketplace.¹¹³

Internalism sets a trap for those concerned with distributive justice in information policy, requiring the distributive impulse to stay within narrow bounds or to become self-defeating. An externalist approach allows us to look more broadly and to see that commons-based and government procurement mechanisms may have constitutive advantages over IP from a distributive perspective.

As Benkler has argued with regard to the commons, justice provides an argument for organizing "a substantial component of our communications and information environment as a commons, in which nonmarket production can take on a more important role."¹¹⁴ This is because the commons may be as efficient as the market at producing information goods (such as encyclopedias),¹¹⁵ and because it seems to have inherent distributive advantages. It will have allocative advantages for the poor because goods produced in the commons

111. This point is well made in Fisher & Syed, *supra* note 65, at 8–10.

112. One can imagine a range of reasons why this might be so. For example, the poor might have educational disadvantages that require tailored informational products, or they might have especially acute needs for information about the distribution of wealth or the functioning of the social welfare state. The poor might also have a special interest in the production of class analysis about, for example, the structure of education or the possibilities of political mobilization. See, e.g., PAULO FREIRE, *PEDAGOGY OF THE OPPRESSED* (Myra Bergman Ramos trans., Continuum Pub. Co. 1990) (1968); FRANCES FOX PIVEN & RICHARD A. CLOWARD, *POOR PEOPLE'S MOVEMENTS: WHY THEY SUCCEED, HOW THEY FAIL* (1979).

113. This does not contradict the internalist point made earlier, that sometimes the poor have more to gain in access than they have to lose in innovation. Precisely because the poor do not have the same ability to pay as the rich, large numbers of the global poor can be given access to, say, HIV medicines, without substantially undermining the potential private market for treatments for HIV.

114. Yochai Benkler, *Freedom in the Commons: Towards a Political Economy of Information*, 52 DUKE L.J. 1245, 1269 (2003); see also BENKLER, *supra* note 18, at 303–06 (arguing, in relatively brief form, that distributive values should lead us to sometimes prefer commons-based schemes of information protection).

115. Benkler, *supra* note 114, at 1270; cf. Jim Giles, *Internet Encyclopaedias Go Head to Head*, 438 NATURE 900 (2005) (finding that peer reviewers identified almost as many errors in science entries in the *Encyclopedia Britannica* as they identified in Wikipedia).

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are made available freely to others. It may also have relative advantages with respect to dynamic production because it allocates and produces resources via signals that are not closely tied to existing distributions of wealth.¹¹⁶

Open and collaborative systems also require resource inputs, but they mobilize these resources in a different fashion than does the IP approach. Some such systems, like Wikipedia or Seti@home, are sustained by voluntary contributions of time or resources, and are indirectly subsidized by income that such participants glean from elsewhere in their lives.¹¹⁷ Others, such as open-source software projects that emerge from universities, are sustained by salaries paid by the public.¹¹⁸ Because these motivations are not systematically related to consumers' willingness and ability to pay, we may expect them to produce information in a way that is more aligned with what distributive justice requires. I say "may" because whether this will be so depends on what in fact motivates such collaborative projects—a subject about which there is debate.¹¹⁹ If, for example, volunteer-based projects like Wikipedia are motivated by ingroup solidarity, they may do little to provide information goods needed particularly by the poor insofar as the poor are less likely to have the time and resources to participate actively in the communities that generate these goods. If these volunteer-based projects are motivated by attempts to benefit from reputational advantages, then their dynamic implications in distributive terms are still less clear. Examples of successful open and collaborative innovation projects are also somewhat limited. It may be possible to create a volunteer-driven encyclopedia online, but more difficult to develop new medicines in peer-to-peer fashion.

In distributive terms, the most promising of the alternatives or supplements to the price mechanism is the approach of government procurement. Government-sponsored prizes and government contracting facilitate access for

116. Benkler, *supra* note 114, at 1271.
 117. For a description of Seti@home, see BENKLER, *supra* note 18, at 81–83.
 118. See generally STEVEN WEBER, THE SUCCESS OF OPEN SOURCE (2004) (providing a history of the open-source software movement).
 119. Compare BENKLER, *supra* note 18, at 92–127 (arguing that collaborative projects arise from networked communities whose motivations often fall outside of rational economic behaviors), and Scott Stern, *Do Scientists Pay to Be Scientists?*, 50 MGMT. SCI. 835, 836 (2004) (analyzing empirical data about wages of scientists with multiple job offers, and concluding that scientists have a "taste" for the scientific ethos that they are willing to pay for with lesser compensation), with Josh Lerner & Jean Tirole, *Some Simple Economics of Open Source*, 50 J. INDUS. ECON. 197, 218 (2002) (emphasizing the role of reputational benefits and concluding that "while some of [the] benefits conferred from participation in open source projects may be less concrete in nature, there also appear [to] be quite tangible—if delayed—rewards").

the poor because the information goods that these approaches produce are classically not priced. In dynamic terms, these approaches also can be directly organized to meet distributive goals and can be funded through progressive taxation.¹²⁰ While one could also address the distributive costs of price by redistributing resources until an equitable allocation is achieved and by allowing IP to do its work thereafter, this is not a simple process, nor would it obviously be more efficient than simply using mechanisms like prizes. In the transnational domain, there is no mechanism for taxes and transfers of this sort, which helps to explain why prizes have been such a dominant part of the conversation about how to allocate more resources toward diseases that disproportionately impact the global poor.¹²¹ In fact, one of the virtues of prizes is that one can implement them without correcting the market for a good first. This may be particularly important in areas with intractable externalities and dim political prospects of market reform, such as climate change.¹²²

It may be easier to see government's possible role in the effective production of science than of culture.¹²³ Yet one of the leading proposals to reorganize the production and distribution of music in light of the new potential for cheap distribution and reuse via digital networks, offered by William Fisher, recommends that we stimulate production not with price but rather with general taxation funds distributed to creators according to their share in online music consumption—music which in turn would be free to users.¹²⁴ This is an inducement prize, in which the size of the prize is determined by measuring the distribution of the good *ex post*. Fisher makes a strong case that such a scheme would be superior in efficiency terms to one that uses price as a signal, and a scheme such as this would also clearly have salutary distributive effects as long as the government tax operated more progressively than price.¹²⁵

Moreover, some information goods are of such foundational importance to human freedoms and capabilities that we believe that everyone should have them regardless of whether they would express a preference for them in market

120. While government taxes also imply some deadweight loss, the deadweight loss associated with IP is expected to be greater. *See supra* note 62.

121. *See* Love & Hubbard, *supra* note 61; Arjun Jayadev & Joseph Stiglitz, *Two Ideas to Increase Innovation and Reduce Pharmaceutical Costs and Prices*, 28 HEALTH AFFAIRS w165 (2008).

122. *See, e.g.*, Adler, *supra* note 19, at 4, 41–42 (explaining that prizes are an important tool to promote innovation particularly in the context of the failures of attempts to properly price carbon).

123. After all, government already provides substantial support for the development of science and technology. *See, e.g.*, BOROUSH, *supra* note 1.

124. FISHER, *supra* note 18, at 202.

125. *See id.* at 216 (describing the operation of such a system, and assuming it would be based upon the existing progressive tax system).

settings. Justice may demand that individuals are provided with education or healthcare, for example, whether or not they would choose to purchase these goods in a market setting. Similarly, distributive justice may provide grounds to supply a person with access to news channels even if, in a market context, they would demonstrate a preference for reality television. We can call information goods that correspond to these capabilities or freedoms “basic” information goods.

Many prominent theories of distributive justice conclude that individuals have entitlements that are independent of what they prefer. Amartya Sen, for example, developed his capabilities-based theory of justice largely through a critique of preference satisfaction as an appropriate measure of justice.¹²⁶ As Sen points out, preferences may be endogenous, or maladaptive, responding more to a history of depredation than to any properly moral account of what individuals deserve in a just society.¹²⁷ John Rawls’s theory of justice similarly addresses itself not to individual preferences, but to “primary goods,” which he argues should be allocated in such a way as to advance the lot of the worst off.¹²⁸ Moreover, our societal judgment about the importance of free primary and secondary education, or, for example, our unwillingness to let people sell themselves or parts of their bodies, suggest that we believe that individuals have entitlements to certain basic goods irrespective of their preferences.¹²⁹

Even in a world with no morally significant distinctions in wealth, then, we might conclude that all individuals should have access to certain basic information goods. When this is the case, IP is a problematic way of producing information not only because price rations access but, more fundamentally,

126. See Amartya Sen, *Equality of What?*, in 1 THE TANNER LECTURES ON HUMAN VALUES 195, 218 (Sterling McMurrin ed., 1980) [hereinafter Sen, *Equality of What?*]; SEN, *supra* note 97.

127. Sen, *Equality of What?*, *supra* note 126, at 203. Sen famously invokes the example of people with disabilities: “[I]f person A as a cripple gets half the utility that the pleasure-wizard person B does from any given level of income,” he notes, “then in the pure distribution problem between A and B the utilitarian would end up giving the pleasure-wizard B more income than the cripple A.” *Id.* The argument that we must affirmatively support the production and distribution of particularly important informational goods could also be made on market failure grounds, cf. C. EDWIN BAKER, MEDIA, MARKETS, AND DEMOCRACY 3–95 (2002), but because this argument can be collapsed into an efficiency account, I leave it aside here.

128. JOHN RAWLS, A THEORY OF JUSTICE 75–76 (1971). For John Rawls, “primary goods” are goods that no rational person would want to do without, such as liberties, income, and self-respect. *Id.* at 62, 90–95.

129. Molly Van Houweling effectively suggests that creative expression itself should be treated as a foundational human capability, and thus it should not be allocated via preferences expressed in markets. See Van Houweling, *supra* note 89, at 1577–78 (“[W]e do not value speech and creativity merely for the utility they generate for willing buyers; so it is not ideal to allocate speech opportunities via the market and would not be ideal even if everyone had equal purchasing power.”).

because price links production to the expression of preferences in markets. From a Demsetzian perspective, IP is preferable to government procurement because entrepreneurs, interpreting market signals, are able to best approximate the social value of a product. But by definition, with basic information goods we want to sever the relationship between market preferences and production. So here, the central virtue of IP is no virtue at all.

The price mechanism could be retrofitted to address this tension that it creates with justice in the production of basic information goods. For example, we might seek to subsidize purchases, and thus the production of basic information goods for the poor, by providing means-tested vouchers for the purchase of particular information goods.¹³⁰ Douglas Lichtman has made such a proposal in the area of medicines, suggesting that the government provide the poor not with income (which could be used to satisfy a variety of preferences), but rather with vouchers to supplement the market demand that the poor can manifest for medicines.¹³¹

However, there are two substantial difficulties with this approach. First, the government would have to determine how large the vouchers should be—that is, it would have to somehow discern the true demand function of each poor person for drugs like Prozac (for depression) or Advair (for asthma). To call such calculations difficult is to understate the matter. Second, precisely because the voucher system requires this judgment, it compromises the degree to which price can serve as a signal that rationalizes and optimizes production and distribution, as it does in the conventional account. Individuals, in disaggregated fashion, do not drive demand. Rather, the decision is made by the size of the vouchers, which is determined by the government. To the extent that we modify price to compensate for the fact that it measures the wrong things from the perspective of distributive justice, we undermine the accorded virtue of price as a signal. It is not at all obvious why we would want to continue to use price when we have already conceded that we must use means other than the market to establish the value of the good in question. In that case, we incur the efficiency costs of price—including both deadweight loss and the risk of racing—but not

130. As noted earlier, we could also simply provide means-tested redistribution of resources, so that the rich and the poor no longer experience morally relevant differences in their ability to pay. That would make price a far more attractive mechanism from a distributive point of view, but it would not sever the relationship between price and preferences, and so it would fail to address the fact that we might want to subsidize particular informational goods. Vouchers can address this, but they run into the problems discussed next in this Part.

131. Douglas Gary Lichtman, *Pricing Prozac: Why the Government Should Subsidize the Purchase of Patented Pharmaceuticals*, 11 HARV. J.L. & TECH. 123 (1997).

its main efficiency benefit. Vouchers might address the distributive concern, but in a manner less efficient than would more direct modes of government procurement. Because efficiency can have distributive implications—for example, it can reduce the resources available to serve the needs of the poor—this is both a distributive and an efficiency concern.

Price cannot both achieve its promise as a signal and meet the demands of distributive justice where basic information goods are concerned. Price may also be distributively problematic where nonbasic information goods are concerned if the background distribution of resources is unjust. Assume, for example, that popular music is a nonbasic information good. It does not follow that there is no distributive problem with directing its production and distribution through the price mechanism. If music is a good—even if it is a nonbasic good—a case can be made on distributive grounds that we should direct its production and distribution in a way that is not systematically sensitive to the underlying distribution of wealth, when that distribution is itself unjust, and when alternative systems exact no toll in efficiency terms.¹³² A prize system of the sort that Fisher has proposed thus has distributive appeal, even if we do not consider music to be a basic information good and even though it directs the production of music through the decentralized expression of preferences.¹³³

An externalist perspective allows distributive justice inquiries a far greater scope than does an internalist perspective. It allows us to see that government procurement and the commons (and especially the former) are institutional approaches with inherent distributive advantages because they sever the links both between ability to pay and production, and when they concern basic information goods, between preference and production. This is not to say that there is one ideal framework for information production from the perspective of distributive justice, or that price could never be the best system from this perspective. But only once we adopt an externalist perspective can we begin to examine the tradeoffs in particular cases. In some areas, government sponsorship

132. Scotchmer and Nancy Gallini seem to come to the opposite conclusion, arguing that price is a mechanism that precisely serves distributive ends, because it puts the costs of innovation on users, instead of on taxpayers who “might rightfully revolt if asked to bear the costs of developing, say, computer games.” Gallini & Scotchmer, *supra* note 46, at 55. But this intuition may be funded by the sense that video games are not goods at all, but rather are “bads.” If we can address underlying unjust inequities in access to resources with a more just system of developing and distributing information—without compromising efficiency—then distributive justice may provide a general argument for doing so.

133. Because Fisher’s scheme would track digital music consumption, it would be sensitive to preference. But it would prevent preference from being systematically influenced by price because the music would be free. FISHER, *supra* note 18, at 202.

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of information production may be in tension with other goods that are important from a distributive perspective, such as other “primary” goods in the Rawlsian scheme or goods important to capabilities according to Sen. For example, government provisioning of political news is plausibly in tension with values of free speech. Whether and when this will be the case and how we should make tradeoffs between, say, the risk of government interference and the possibility of better meeting the informational needs of the poor are complex questions that require contextual analysis. My aim here is not to adjudicate the debate between mechanisms in distributive terms, but rather to make clear that this debate is one very much worth having. And it can only be had if we first broaden our perspective to bring approaches beyond IP more centrally into view.

III. INTELLECTUAL PROPERTY EXTERNALISM AND THE COST OF PRICE TO INFORMATIONAL PRIVACY

An externalist view can also yield new insights into the relationship between different institutional approaches to cultural and scientific production and informational privacy—insights that are not visible from within an internalist account. Taking an externalist view allows us to ask not only whether we should adjust IP law in some way to accommodate privacy interests,¹³⁴ but also whether some approaches to scientific and cultural production are constitutively more likely than others to conflict with the desire to protect informational privacy.

Asking the question this way makes it possible to see that IP, as a strategy, imposes a cost on privacy—a cost that other institutional approaches do not impose. As I show in this Part, choosing IP as a means to produce information sets up a certain logic: The imposition of price triggers the problem of deadweight loss, and in response we call for price discrimination—that is, for sellers of information goods not just to set one or a few price tiers, but to charge each individual user according to her willingness and ability to pay. We want

134. For important work in this vein, see, for example, Julie E. Cohen, *Copyright and the Jurisprudence of Self-Help*, 13 BERKELEY TECH. L.J. 1089 (1998) (invoking privacy values, among others, in justifying a “right to hack,” that is, to breach digital fences to exercise entitlements granted by copyright law); Julie E. Cohen, *A Right to Read Anonymously: A Closer Look at “Copyright Management” in Cyberspace*, 28 CONN. L. REV. 981 (1996) [hereinafter Cohen, *Right to Read*] (describing conflicts between privacy values and digital rights management, and arguing that the First Amendment should be read to create limits on the scope of digital rights management); Sonia K. Katyal, *Privacy v. Piracy*, 7 YALE J.L. & TECH. 222 (2004–2005) (drawing attention to conflicts between online copyright enforcement efforts and privacy, and arguing for, among other things, changes to the Digital Millennium Copyright Act’s enforcement provisions to accommodate privacy interests).